

NeuralForge

A Feedforward Neural Network Engine

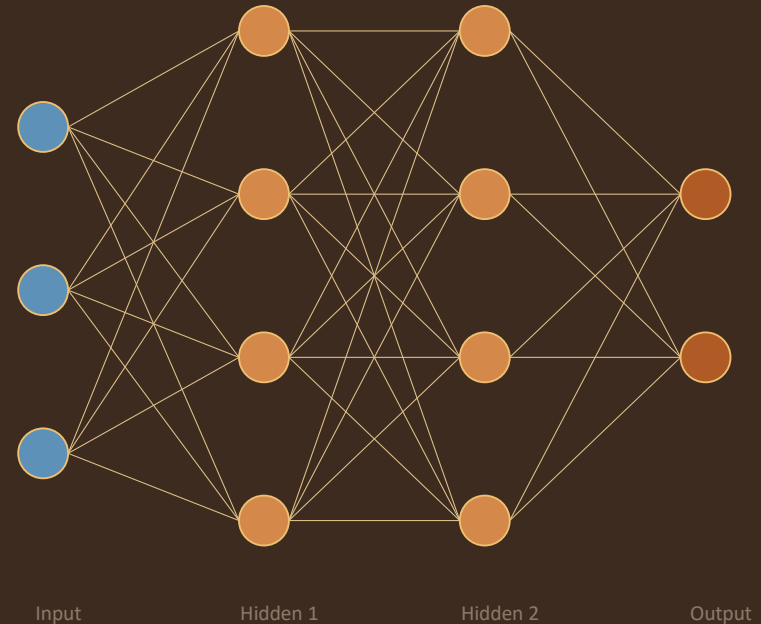
Built from scratch in Java · No ML dependencies

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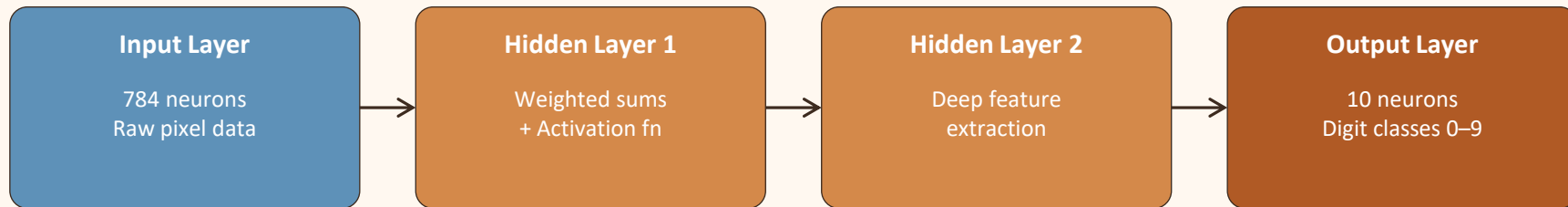
Roll no. : 1720

Supervisor : Dr. Emon Kumar Dey

Course : Software Project Lab 1



What is NeuralForge?



Core Highlights

- ◆ Built entirely from scratch in Java
- ◆ No external ML dependencies
- ◆ Custom matrix library at the core
- ◆ Trains on MNIST — 60,000 handwritten digits
- ◆ Configurable layer depth & neuron counts
- ◆ GUI: draw a digit → live classification

Key Features



Configurable Architecture

Layers & neurons
set via config file



Forward & Backpropagation

Chain rule gradient
computation



Activation Functions

Sigmoid · ReLU
Softmax per layer



Loss Functions

MSE and
Cross-Entropy



Mini-Batch SGD

Configurable batch
size & learn rate



Model Save/Load

Serialize trained
weights to disk



MNIST Training

60,000 images · Loss & accuracy tracked per epoch



Live GUI Canvas

Draw a digit with mouse → real-time confidence scores

Algorithms & Mathematics

01 Matrix Operations

$$A = f (W \cdot X + b)$$

Custom matrix library
Multiplication · Transpose
Element-wise operations

02 Feedforward Propagation

$$A[l] = f (W[l] \cdot A[l-1] + b[l])$$

Sequential layer activation
Output left-to-right
Xavier weight initialisation

03 Backpropagation

$$dL/dW \text{ via Chain Rule}$$

Error gradient flows backward
Each weight's contribution
computed layer by layer

04 Mini-Batch SGD

$$W \leftarrow W - \alpha * dW$$

Batch-averaged gradients
Configurable alpha & batch size
Faster than full-batch GD

Tools & Technologies



Java

- ◆ Object-oriented design
- ◆ No external ML libraries
- ◆ Custom matrix engine from scratch



Java Swing

- ◆ GUI digit-drawing canvas
- ◆ Real-time confidence scores
- ◆ Mouse-based draw input



Git / GitHub

- ◆ Incremental commit history
- ◆ Algorithm-by-algorithm progress
- ◆ Version control & backup

Thank You